

Effect of Lambda Plot

Simulate student t data with $n = 100, 1000, 10000$ and d.o.f. = 5,10,20,100

```
library(tidyverse)
library(rstan)
```

```
set.seed(454)
samples1 <- rt(1000, df = 10)
samples <- as.data.frame(samples1)

names <- c("mean", "se_mean", "sd", "2.5%", "25%", "50%", "75%",
           "97.5%", "n_eff", "Rhat") #column names
```

```
#Exp(1/5)

exp1_data <- list(N = nrow(samples),
                 y = samples$samples1,
                 rate = 1/5)

exp1.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  exp1_fit <- stan(file = "fig2_run1.stan",
                  data = exp1_data,
                  chains = 4,
                  warmup = 1000,
                  iter = 2000)

  a<- as.list(summary(exp1_fit, pars = "nu")$summary)
```

```

    exp1.df[i,] <- a
  }

  colnames(exp1.df) <- names

```

```

#Exp(1/10)

exp2_data <- list(N = nrow(samples),
                  y = samples$samples1,
                  rate = 1/10)

exp2.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  exp2_fit <- stan(file = "fig2_run1.stan",
                  data = exp2_data,
                  chains = 4,
                  warmup = 1000,
                  iter = 2000)

  a<- as.list(summary(exp2_fit, pars = "nu")$summary)
  exp2.df[i,] <- a
}

colnames(exp2.df) <- names

```

```

#Exp(1/20)

exp3_data <- list(N = nrow(samples),
                  y = samples$samples1,
                  rate = 1/20)

exp3.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  exp3_fit <- stan(file = "fig2_run1.stan",
                  data = exp3_data,
                  chains = 4,
                  warmup = 1000,
                  iter = 2000)

  a<- as.list(summary(exp3_fit, pars = "nu")$summary)

```

```

    exp3.df[i,] <- a
  }

  colnames(exp3.df) <- names

```

```

#Exp(1/100)

exp4_data <- list(N = nrow(samples),
                  y = samples$samples1,
                  rate = 1/100)

exp4.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  exp4_fit <- stan(file = "fig2_run1.stan",
                  data = exp4_data,
                  chains = 4,
                  warmup = 1000,
                  iter = 2000)

  a<- as.list(summary(exp4_fit, pars = "nu")$summary)
  exp4.df[i,] <- a
}

colnames(exp4.df) <- names

```

```

#PC(0.2)

pc1_data <- list(N = nrow(samples),
                  y = samples$samples1,
                  lambda = -log(0.2)/10)

pc1.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))
init_fun <- function() {
  list(nu = 10)
}

for (i in 1:10){
  pc1_fit <- stan(file = "fig2_PCprior.stan",
                  data = pc1_data,
                  init = init_fun,
                  chains = 4,

```

```

        warmup = 1000,
        iter = 2000)

a<- as.list(summary(pc1_fit, pars = "nu")$summary)
pc1.df[i,] <- a
}

colnames(pc1.df) <- names

```

```

#PC(0.3)

pc2_data <- list(N = nrow(samples),
                y = samples$samples1,
                lambda = -log(0.3)/10)

pc2.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  pc2_fit <- stan(file = "fig2_PCprior.stan",
                 data = pc2_data,
                 init = init_fun,
                 chains = 4,
                 warmup = 1000,
                 iter = 2000)

  a<- as.list(summary(pc2_fit, pars = "nu")$summary)
  pc2.df[i,] <- a
}

colnames(pc2.df) <- names

```

```

#PC(0.4)

pc3_data <- list(N = nrow(samples),
                y = samples$samples1,
                lambda = -log(0.4)/10)

pc3.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  pc3_fit <- stan(file = "fig2_PCprior.stan",
                 data = pc3_data,

```

```

        init = init_fun,
        chains = 4,
        warmup = 1000,
        iter = 2000)

a<- as.list(summary(pc3_fit, pars = "nu")$summary)
pc3.df[i,] <- a
}

colnames(pc3.df) <- names

```

```

#PC(0.5)

pc4_data <- list(N = nrow(samples),
                y = samples$samples1,
                lambda = -log(0.5)/10)

pc4.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  pc4_fit <- stan(file = "fig2_PCprior.stan",
                 data = pc4_data,
                 init = init_fun,
                 chains = 4,
                 warmup = 1000,
                 iter = 2000)

  a<- as.list(summary(pc4_fit, pars = "nu")$summary)
  pc4.df[i,] <- a
}

colnames(pc4.df) <- names

```

```

#PC(0.6)

pc5_data <- list(N = nrow(samples),
                y = samples$samples1,
                lambda = -log(0.6)/10)

pc5.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){

```

```

pc5_fit <- stan(file = "fig2_PCprior.stan",
               data = pc5_data,
               init = init_fun,
               chains = 4,
               warmup = 1000,
               iter = 2000)

a<- as.list(summary(pc5_fit, pars = "nu")$summary)
pc5.df[i,] <- a
}

colnames(pc5.df) <- names

```

```

#PC(0.7)

pc6_data <- list(N = nrow(samples),
                 y = samples$samples1,
                 lambda = -log(0.7)/10)

pc6.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

for (i in 1:10){
  pc6_fit <- stan(file = "fig2_PCprior.stan",
                 data = pc6_data,
                 init = init_fun,
                 chains = 4,
                 warmup = 1000,
                 iter = 2000)

  a<- as.list(summary(pc6_fit, pars = "nu")$summary)
  pc6.df[i,] <- a
}

colnames(pc6.df) <- names

```

```

#PC(0.8)

pc7_data <- list(N = nrow(samples),
                 y = samples$samples1,
                 lambda = -log(0.8)/10)

pc7.df <- as.data.frame(matrix(NA, nrow = 10, ncol = 10))

```

```

for (i in 1:10){
  pc7_fit <- stan(file = "fig2_PCprior.stan",
                 data = pc7_data,
                 init = init_fun,
                 chains = 4,
                 warmup = 1000,
                 iter = 2000)

  a<- as.list(summary(pc7_fit, pars = "nu")$summary)
  pc7.df[i,] <- a
}

colnames(pc7.df) <- names

# Combine and Plot them!
frames <- list(exp1.df, exp2.df, exp3.df, exp4.df, pc1.df,pc2.df,pc3.df,pc4.df,pc5.df,pc6.df,pc7.df)

combined <- as.data.frame(matrix(NA, nrow = 11, ncol = 3))
names2 <- c("2.5%", "50%", "97.5%")
colnames(combined) <- names2

combined[1,] <- c(mean(exp1.df$`2.5%`), mean(exp1.df$`50%`),mean(exp1.df$`97.5%`))
combined[2,] <- c(mean(exp2.df$`2.5%`), mean(exp2.df$`50%`),mean(exp2.df$`97.5%`))
combined[3,] <- c(mean(exp3.df$`2.5%`), mean(exp3.df$`50%`),mean(exp3.df$`97.5%`))
combined[4,] <- c(mean(exp4.df$`2.5%`), mean(exp4.df$`50%`),mean(exp4.df$`97.5%`))
combined[5,] <- c(mean(pc1.df$`2.5%`), mean(pc1.df$`50%`),mean(pc1.df$`97.5%`))
combined[6,] <- c(mean(pc2.df$`2.5%`), mean(pc2.df$`50%`),mean(pc2.df$`97.5%`))
combined[7,] <- c(mean(pc3.df$`2.5%`), mean(pc3.df$`50%`),mean(pc3.df$`97.5%`))
combined[8,] <- c(mean(pc4.df$`2.5%`), mean(pc4.df$`50%`),mean(pc4.df$`97.5%`))
combined[9,] <- c(mean(pc5.df$`2.5%`), mean(pc5.df$`50%`),mean(pc5.df$`97.5%`))
combined[10,] <- c(mean(pc6.df$`2.5%`), mean(pc6.df$`50%`),mean(pc6.df$`97.5%`))
combined[11,] <- c(mean(pc7.df$`2.5%`), mean(pc7.df$`50%`),mean(pc7.df$`97.5%`))

types <- c("exp(1/5)", "exp(1/10)", "exp(1/20)", "exp(1/100)", "pc(0.2)", "pc(0.3)", "pc(0.4)", "pc(0.5)", "pc(0.6)", "pc(0.7)")
combined <- cbind(combined, types)

class <- c(rep("exp", 4), rep("pc",7))
combined <- cbind(combined, class)

ggplot(combined, aes(x = class, y = `50`, color = types)) +
  geom_pointrange(

```

```
    aes(ymin = `2.5`, ymax = `97.5`),  
    position = position_dodge(width = 0.6)  
  ) +  
  geom_hline(yintercept = 10) +  
  theme_classic() +  
  labs(x = "Prior Class", y = "d.o.f.", color = "Priors", title = "Effect of Lambda")
```